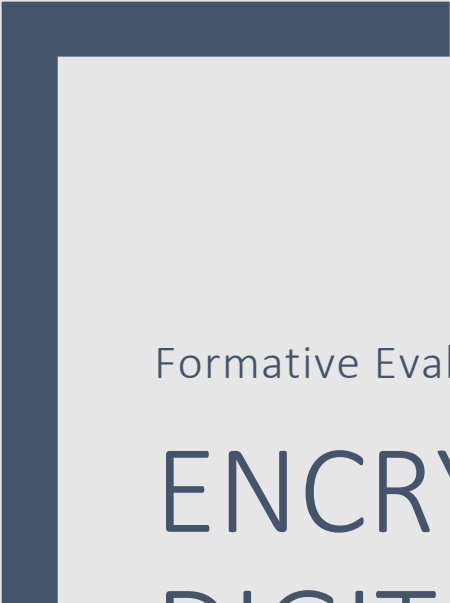




Formative Evaluation Plan

ENCRYPTION FOR DIGITAL IDENTITIES

John McSwain

jmcswain3@sc.edu

EDET 722-J52



Needs Assessment

Introduction

This report outlines the needs analysis and analysis report of an instructional learning unit for encryption for digital identities, highlighting encryption's goals and identifying several encryption types. The target audience for the learning unit includes software application users who expect of privacy and confidentiality for their digital identities. Digital identities contain "...data that uniquely describes a person or thing (called the *subject* or *entity* in the language of digital identity) but also contains information about the subject's relationships to other entities." (Windley, 2005)

Purpose

Encryption includes any method of transforming information into an indiscernible form to ensure confidentiality (Jean-Philippe Aumasson, 2018). Users of application software rely on encryption and decryption to maintain their expectation of digital privacy. While cryptography is essential to secure communications, the mechanics of an application's encryption are reduced marketing terms like "most secure" to gain user trust and brand loyalty. This learning unit will highlight cryptography's functional goals and types to increase user awareness of application security. Specifically, the unit will identify privacy, authentication, integrity, and nonrepudiation (Windley, 2005) as encryption goals. In addition, the unit will introduce the types of encryption, namely symmetric, asymmetric, and hashes, to help users properly consider encryption for their digital identity management.

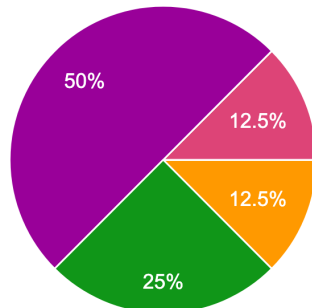
Process

The needs assessment featured Google Forms Survey sent to anonymous users of web-based software applications (e.g., Google Chrome, Mozilla Firefox, Zoom, Slack, Grammarly, Dropbox, LinkedIn, Facebook, Cash App, etc.). The purpose of the survey was to assess users' understanding of the purpose, value, and methodology of encryption in the applications they regularly use. The survey was made up of 12 close-ended questions: Questions 1-3 (demographic), Questions 4-5 (application use), Questions 6-12 (security and encryption awareness). The survey's needs types included critical incident and normative questions regarding usage and privacy expectations. In addition, several questions in the security and awareness section used a Likert style scale to measure respondents' comprehension of the importance and mechanics of encryption.

Survey Results

What is your current age range?

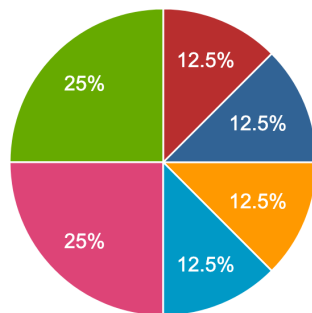
8 responses



- Under 12 years old
- 12-17 years old
- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65-74 years old
- 75 years or older

What is the highest degree or level of school you have completed? If currently enrolled, highest degree received.

8 responses

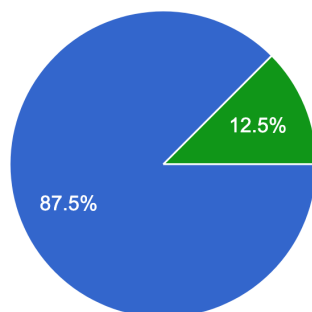


- Nursery school to 8th grade
- Some high school, no diploma
- High school graduate, diploma or the...
- Some college credit, no degree
- Trade/technical/vocational training
- Associate degree
- Bachelor's degree
- Master's degree

▲ 1/2 ▼

Are you currently...?

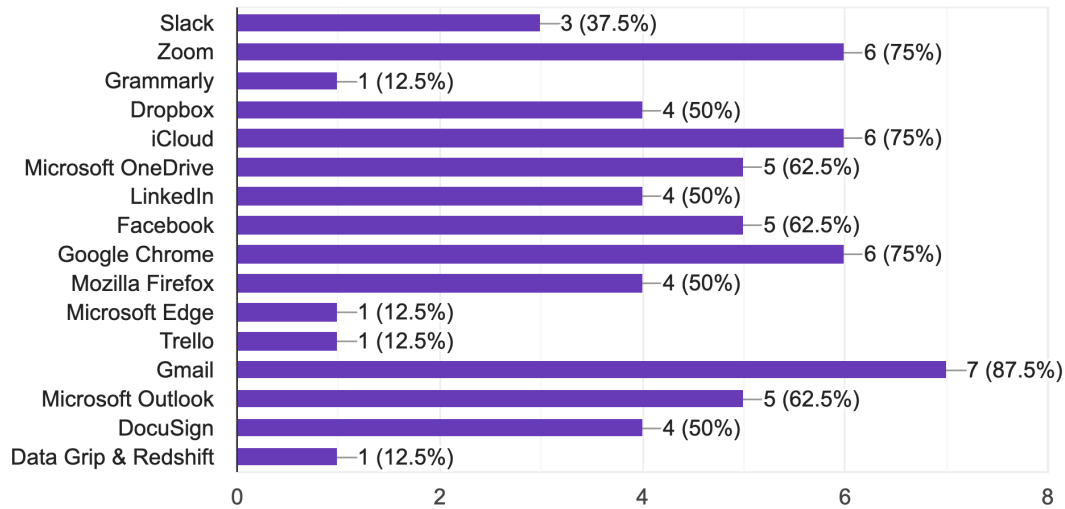
8 responses



- Employed
- Self-employed
- Out of work and looking for work
- Out of work but not currently looking for work
- A homemaker
- A student
- Military
- Retired
- Volunteer

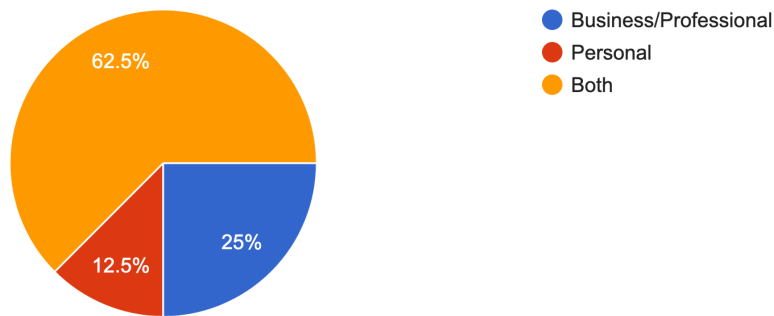
Do you use any of the web or desktop software applications listed below? Please select all that apply.

8 responses



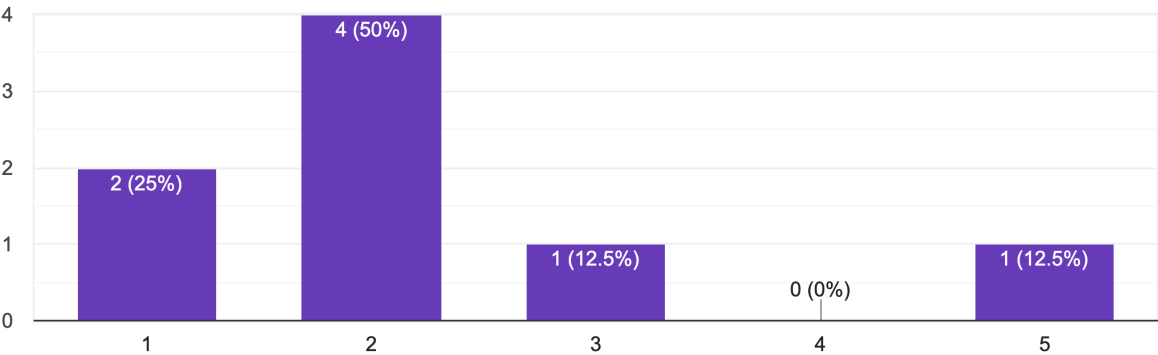
What is the purpose of using the software apps you selected above?

8 responses



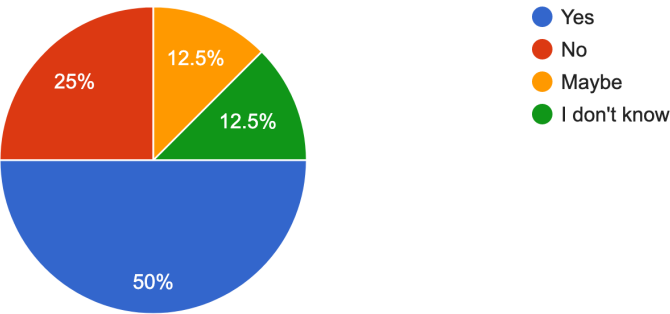
How confident do you feel about your understanding of how web or desktop software applications work?

8 responses



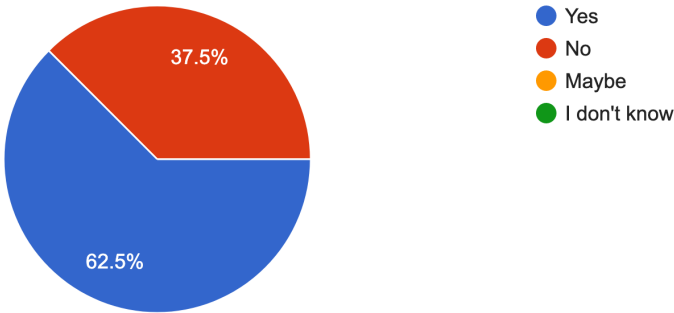
Do you consider digital privacy when choosing or using a software application?

8 responses



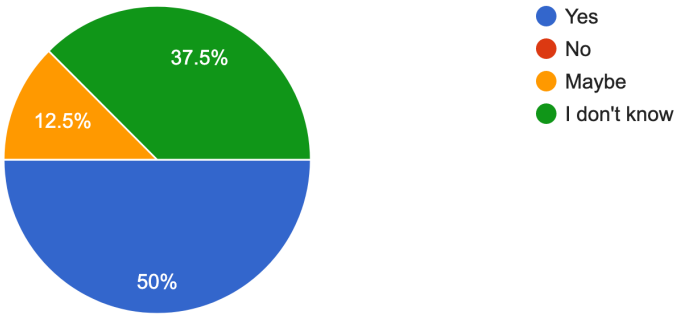
Have you ever been notified of a security incident/breach with any web or desktop applications you use?

8 responses



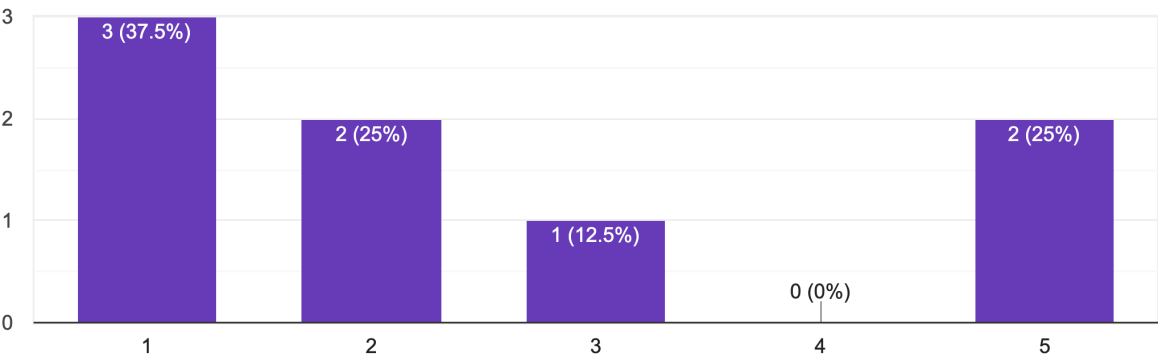
Do you know if any software applications you selected use encryption for data in use, in motion, or at rest?

8 responses



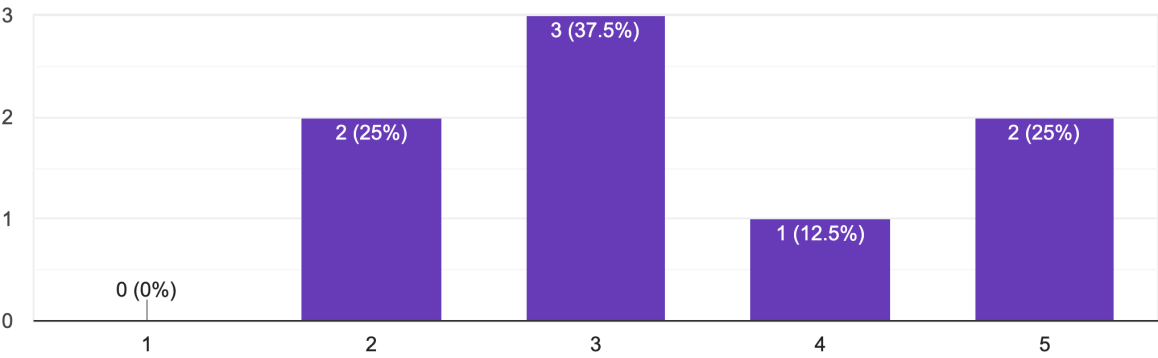
How confident do you feel about your understanding of the importance of encryption?

8 responses



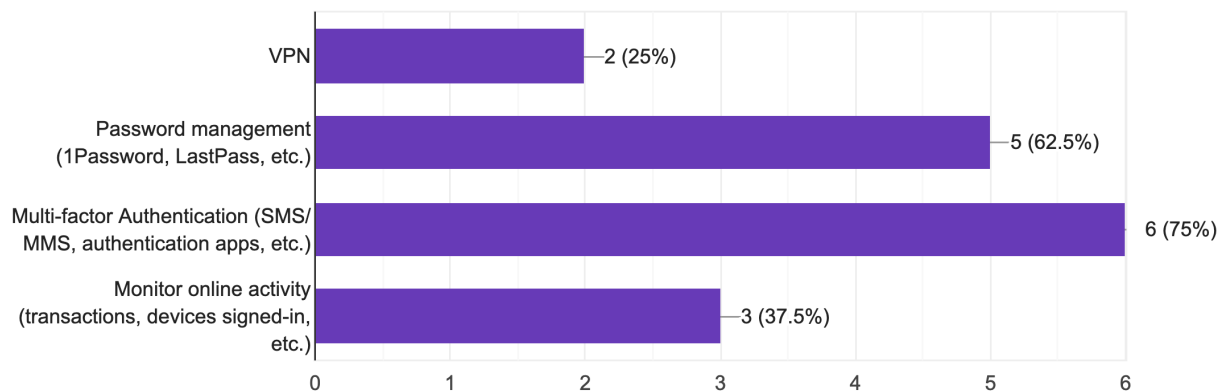
How confident do you feel about your understanding of how encryption works?

8 responses



Do you take any extra steps to protect your digital identity? Please select all that apply.

8 responses



Needs Assessment Recommendations/Action

Many users use a variety of software applications for work and personal use. While 75% (6 out of 8 scored 1 to 3) of respondents claim to be aware of the importance of encryption, 75% (6 out of 8 scored 3 through 5) of respondents lack confidence in their knowledge of how encryption works. All (100%) respondents use some form of security measure to protect their digital identity, and 37.5% of respondents don't know if their apps use encryption. Based on this data, the instructional unit will describe how encryption works to enlighten users who rely heavily on software applications for protection. In addition, the instructional unit will include examples of digital information and communication which can be encrypted and decrypted. Finally, the learning unit will consist of examples of widely used applications (e.g., Dropbox, Slack) and their encryption methods.

Analysis Report

Goal Analysis

The learning unit aims to increase user awareness and understanding of how and where software applications use encryption concerning users' digital identities.

Initial List of Goals

- Define a digital identity and privacy
- Define and describe authentication, identification, nonrepudiation
- Distinguish between plaintext and cipher text
- Define encryption
- Define symmetrical encryption
- Define asymmetrical encryption
- Define hash encryption

Refinement of goals

- Understand the application of identity, security, and privacy
- Understand the process of encryption and decryption
- Understand the value of authentication, identification, nonrepudiation
- Distinguish between asymmetrical, symmetrical, and hash encryption

Rank goals

- Understand the application of identity, security, and privacy
- Understand the process of encryption and decryption
- Understand the value of authentication, identification, nonrepudiation
- Distinguish between asymmetrical, symmetrical, and hash encryption

Second Refinement of goals

- Understand a “digital identity” and user’s expectation of security and privacy
- Understand encryption’s role of providing privacy, authentication, integrity, nonrepudiation to digital identities
- Understand the process of encryption and decryption
- Distinguish between asymmetrical, symmetrical, and hash encryption

Final ranking

- Understand a “digital identity” and user’s expectation of security and privacy
- Understand encryption’s role of providing privacy, authentication, integrity, nonrepudiation to digital identities
- Understand the process of encryption and decryption
- Distinguish between asymmetrical, symmetrical, and hash encryption

Learner Analysis

General Characteristics

- 75% of the target audience will be users aged between 35-44 years old
- 100% of the target audience has a high school diploma with 75% completing a Bachelor’s degree or higher
- 87.5% of the target audience is employed
- 75% of respondents use Zoom, iCloud, Google Chrome, and Gmail

Special Entry Characteristics and Prior Knowledge

- Basic computer literacy
- Fluent in U.S. English
- Understanding of basic arithmetic

Additional Important Characteristics

- Understanding of privacy and confidentiality

Contextual Analysis

Orienting context

Learners for this unit belong to an employed, computer literate segment of software application users who do not know if/how their applications protect their data or privacy. A learning unit outlining the goals and types of encryption can provide users with an awareness that can help them properly scrutinize applications that may put their digital identities at risk.

Instructional context

Learners will run a self-paced Microsoft PowerPoint 15 minute "Encryption for Digital identities" instructional unit on a Windows/Mac/Linux laptop (minimum 13" screen) or desktop (24" + screen recommended) with a screen aspect ratio of 16:9. The unit will contain both the goals of encryption and three types of encryption.

Transfer context/Performance context

Learners can complete this unit during their continued personal or business use of web or desktop application software. Following completion of the module, users should be able to ask and search for specific cryptographic information about how their data is protected.

Topical Task Analysis

1. Goal: Understand a “digital identity” and user’s expectation of security and privacy

1.1. Terminology

- 1.1.1. Digital Identity – tiers of data about a subject (person, place, thing) that represent attributes, preferences, and traits
- 1.1.2. Identity management – policies to create, manage, and enforce a digital identity roles and permissions
- 1.1.3. Security – absence of harm or threat (destruction, tampering, unwanted visibility)
- 1.1.4. Privacy – protection and control of access to the traits and characteristics of a digital identity
- 1.1.5. Trust – belief in the good faith or integrity of another identity
- 1.1.6. Encryption – process of making plain information incomprehensible to observers and only readable by the intended sender(s) or receiver(s) who may possess a secret key
- 1.1.7. Decryption – process of making incomprehensible information readable using a secret
- 1.1.8. Nonrepudiation – existence of unforgeable evidence information was sent or received
- 1.1.9. Algorithm – a list of procedural steps
- 1.1.10. Ciphertext – encrypted information
- 1.1.11. Plaintext – unencrypted information

1.1.12. Digital certificate – digital file issued by a 3rd party that proves the authenticity of a digital identity and establish a trusted, secure connection or channel of communication

1.1.13. Digital signature (hash) – digital file which accompanies a message to validate authenticity and integrity of the message

1.1.14. Checksum – series of numbers and letters which represent an original

1.1.15. Hash – algorithm that converts any length of input and produces a fixed size output (e.g., MD5, SHA256)

1.1.16. Key – a secret value used with an encryption algorithm to encrypt and decrypt information

1.2. Role of digital identities

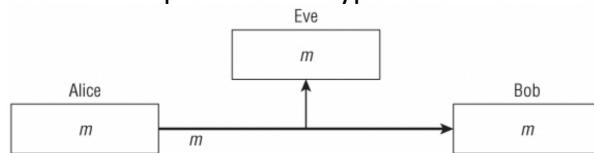
1.2.1. Authentication – using traits or characteristics (factors) to verify a digital identity

1.2.2. Authorization – using digital identities to determine a digital identity's level of access

1.2.3. Pseudonymity – unique identities which cannot be linked to other identities

2. Goal: Understand encryption's role of providing privacy, authentication, integrity, nonrepudiation to digital identities

2.1. What is the problem encryption solves?



How can Alice and Bob communicate securely? (Tadayoshi Kohno et al., 2010)

2.2. Practical examples of the benefits of encryption

2.2.1. Financial transaction (privacy, authentication)

2.2.1.1. Buying a car and verifying identity and financial status

2.2.2. Purchasing medication or alcohol (authentication, authorization)

2.2.2.1. Presenting government issued ID to validate the buyers drinking age

2.2.3. Digital certificates & signatures (nonrepudiation)

2.2.3.1. Visiting google.com vs googl3.com

2.2.3.2. Sending/receiving a private email

2.2.4. Checksum (hash algorithm) (integrity)

2.2.4.1. Downloading Linux image with checksum which matches the origin

2.2.5. VPN (privacy, authentication, authorization)

2.2.5.1. Connecting to an organization intranet

2.3. Threats

2.3.1. The absence of encryption and decryption would mean zero trust, no privacy, no means of verifying integrity, and no mechanism for validating authenticity

2.3.2. Combat digital leakage (loss of digital data)

3. Goal: Understand the process of encryption and decryption

3.1. Convert plaintext into ciphertext using a basic encryption process called the Caesar Cipher

3.1.1. Example (Caesar Cipher, key = 5):

Plaintext: Once upon a time not long ago when people wore pajamas and lived life slow

Ciphertext: tshj zuts f ynrj sty qtsl flt bmjs ujtuqj btwj ufofrfx fsi qnaji qnkj xqtb

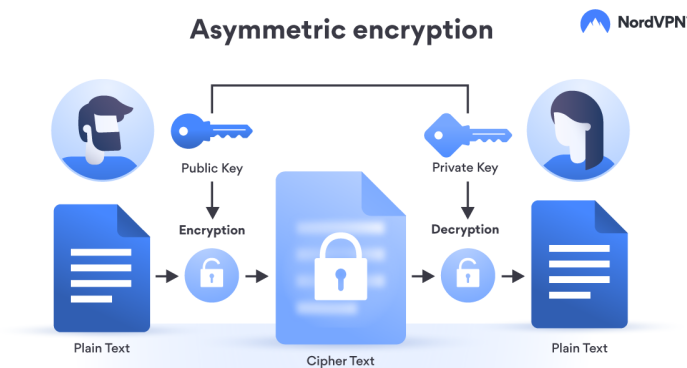
3.1.2. Kerckhoff's principle - security of a cipher should rely only on the secrecy of the key and not on the secrecy of the cipher.

3.1.3. Encryption strength is a function of key size

4. Goal: Distinguish between asymmetrical, symmetrical, and hash encryption

4.1. Asymmetrical (public-key) encryption uses two keys: public key and private key combination (not equal) to encrypt and decrypt

Example



(What Is Asymmetric Cryptography? | NordVPN, 2020)

4.2. Symmetrical-key encryption uses the same key to encrypt and decrypt

4.2.1. Example (AES, ECB, key = EnXnG6Km!z7urXrx, encoding = Base64) (DevGlan, n.d.)

4.2.1.1. Plaintext:

I wanna rock right now
I'm Rob Base and I came to get down
I'm not internationally known
But I'm known to rock the microphone

Ciphertext:

WUDMzGTeOrHTw/5q7OKO62AUx4gf3OJoo7qeiLmrG+5y9BMudvJTQWP
6K21rTImB4dVki1atAsm7tlw4tR3f+qLkop9op5AyYdvk+4u789U6wc/RJjH3
J/nSYT+BG3v8BCIT2pbeEZeN9LFCf9WIlfg0NMbx/31ECAWG++Cu+M=

4.3. Hash functions are one-way functions whereas data of any size is converted into a fixed result and theoretically cannot be decrypted.

4.3.1. Example (SHA256)

4.3.1.1. Plaintext:

All right!
Stop whatcha doin'

'Cause I'm about to ruin
The image and the style that ya used to
I look funny
But yo I'm makin' money, see
So yo world I hope you're ready for me
Now gather round
I'm the new fool in town
And my sound's laid down by the Underground
I drink up all the Hennessey ya got on ya shelf
So just let me introduce myself
Ciphertext:
e8a37f6b31f6918cc597672c3dd30d53f8b7516bb56634e9df15684b8a03f
40e

Performance Objectives and Assessment Items

Final Instructional Goals

1. Understand a “digital identity” and user’s expectation of security and privacy

Performance Objective	Classification	Assessment Items
Given a digital identity, the learner will be able to identify the conditions needed to establish trust in private and security communication	Principles and rules - Recall	T/F – When communicating private information over the Internet, encoding the information is the best way to ensure the information remains private and secure. Key: F
		T/F – When communicating private information over the Internet, there is no way to verify if the intended recipient is genuine vs an imposter. Key: F

2. Understand encryption’s role of providing privacy, authentication, integrity, nonrepudiation to digital identities

Performance Objective	Classification	Assessment Items
Given a situation where digital identities are communicating, the learner will be able to distinguish	Principles and rules - Recall	Which website appears to use an encrypted connection? a. http://example.com

between encrypted and unencrypted communication		b. http://sc.edu c. http://google.com d. https://eric.ed.gov/ Key: d
		What kind of fingerprint can be used to ensure the integrity of digital information has not been altered? a. digital certificate b. vpn c. checksum d. encryption Key: c

3. Understand the process of encryption and decryption

Performance Objective	Classification	Assessment Items
Given plaintext, the learner will be able to encrypt plaintext into ciphertext and decrypt ciphertext into plaintext using the Caesar Cipher.	Procedure - Application	Using Caesar Cipher encryption (key/shift of 8), the plaintext below would match which of the ciphertext choices below: “This is a test message” a. Bpqa qa i bmab umaaiom b. Hvwg wg o hsg h asggous c. Pdeo eo w paop iaowca d. Qefp fp x qbpq jbppxdb Key: a
		Using Caesar Cipher decryption, what would the key need to be for the ciphertext to match the plaintext below: L fdq gr dqbwklqj I can do anything a. 9 b. 5 c. 3 d. 1

		Key: c
--	--	--------

4. Distinguish between asymmetrical, symmetrical, and hash encryption

Performance Objective	Classification	Assessment Items
Given a digital identity, the learner will be able to identify the advantages and disadvantages of each encryption type.	Principles and rules - Recall	T/F Hash functions, when used, provide a result which cannot be reversed to determine the original digital information. Key: T
		T/F Asymmetrical encryption uses the same key for encryption and decryption Key: F

Instructional Strategy Plan

Sequencing Strategy

The instructional unit will follow several phenomena from Posner and Strike's learning-related sequencing. The learning unit will use familiarity of common, personal digital information and privacy expectations and use a development strategy for the introduction of encryption and decryption. Finally, a difficulty strategy will be used to describe symmetrical, asymmetrical, and hash encryption "... by the fineness of the discrimination the learner must make, how quickly the procedure is executed, and the amount of cognitive processing required" (Morrison, 2019, p. 131).

Preinstructional Strategy

The preinstructional strategy for this learning unit consists of behavioral objectives, "...often simply a restatement of the objectives the designer has developed or goal statements describing the behavior the student must master (Morrison, 2019, p. 172)", and an overview "... written at the same level of abstraction as the unit of instruction and simply serve to introduce the student to the central themes" (Morrison, 2019, p. 174).

Instructional Sequence and Strategy

Goal: Understand a "digital identity" and user's expectation of security and privacy

Objective: Given a digital identity, the learner will be able to identify the conditions needed to establish trust in private and security communication (principles and rules – recall)

Initial Presentation: A series of key terms and examples will be presented: digital identity, encryption, identity management, security, privacy, and trust. Thereafter, examples of the role of digital identities for authentication and authorization will be presented to associate terms with familiar practices.

Generative Strategy: RUL-EG and integration

Learner Practice Items:

T/F – When communicating private information over the Internet, encoding the information is the best way to ensure the information remains private and secure.

Key: F

T/F – When communicating private information over the Internet, there is no way to verify if the intended recipient is genuine vs an imposter.

Key: F

Goal: Understand encryption's role of providing privacy, authentication, integrity, nonrepudiation to digital identities

Objective: Given a situation where digital identities are communicating, the learner will be able to distinguish between encrypted and unencrypted communication (principles and rules – recall)

Initial Presentation: A series of key terms and examples will be presented: nonrepudiation, authentication, authorization, and integrity. Thereafter, examples and scenarios illustrating the presence and lack of privacy

Generative Strategy: RUL-EG and integration

Learner Practice Items:

Which website appears to use an encrypted connection?

- a. <http://example.com>
- b. <http://sc.edu>
- c. <http://google.com>
- d. <https://eric.ed.gov/>

Key: d

What kind of fingerprint can be used to ensure the integrity of digital information has not been altered?

- a. digital certificate
- b. vpn
- c. checksum
- d. encryption

Key: c

Goal: Understand the process of encryption and decryption

Objective: Given plaintext, the learner will be able to encrypt plaintext into ciphertext and decrypt ciphertext into plaintext using the Caesar Cipher (procedure - application)

Initial Presentation: A description of encryption and decryption will be provided along with a description of a secret key. Thereafter, an example of a Caesar Cipher will be provided, illustrating the shift of letters using a secret key to encrypt and decrypt a message

Generative Strategy: RUL-EG and organization

Learner Practice Items:

Using Caesar Cipher encryption (key/shift of 8), the plaintext below would match which of the ciphertext choices below:

"This is a test message"

- a. Bpqa qa i bmab umaaiom
- b. Hvwg wg o hsg h asggous
- c. Pdeo eo w paop iaowca
- d. Qefp fp x qbpq jbppxdb

Key: a

Using Caesar Cipher decryption, what would the key need to be for the ciphertext to match the plaintext below:

L fdq gr dqbwklqj

I can do anything

- a. 9
- b. 5
- c. 3
- d. 1

Key: c

Goal: Distinguish between asymmetrical, symmetrical, and hash encryption

Objective: Given a digital identity, the learner will be able to identify the advantages and disadvantages of each encryption type. (principles and rules – recall)

Initial Presentation: A table describing each term, definition, mechanics, and use cases will be provided for context as to when each type of encryption should be used.

Generative Strategy: RUL-EG and organization

Learner Practice Items:

T/F Hash functions, when used, provide a result which cannot be reversed to determine the original digital information.

Key: T

T/F Asymmetrical encryption uses the same key for encryption and decryption

Key: F

Design Plan

Design Sequence	Description	Objective	Estimated Time	Instructional Strategy
Preinstructional Strategy	Overview Objectives	---	2-3 minutes	---
Instruction for Goal 1	Identify the conditions needed to establish trust in private and secure communication	1.1	3-5 minutes	Initial Presentation: terms, concepts, examples Generative Strategy: RUL-EG and integration Learner Practice
Instruction for Goal 2	Distinguish between encrypted and unencrypted communication	2.1	3-5 minutes	Initial Presentation: terms, concepts, examples, scenarios Generative Strategy: RUL-EG and integration Learner Practice
Instruction for Goal 3	Encrypt plaintext into ciphertext and decrypt ciphertext into plaintext using the Caesar Cipher.	3.1	3-5 minutes	Initial Presentation: concepts, examples Generative Strategy: RUL-EG and organization Learner Practice
Instruction for Goal 4	Identify the advantages and disadvantages of each encryption type.	4.1	3-5 minutes	Initial Presentation: terms, traits, examples Generative Strategy: RUL-EG and organization Learner Practice
Posttest	Online test in GoogleForms	---	7-10 minutes	---

Formative Evaluation Plan

Purpose

The formative evaluation plan aims to measure the effectiveness of the "Encryption of Digital Identities" learning unit as the content progresses through the stated learning objectives. Results of the evaluation aid in updating content, messaging, and delivery where applicable and assess learner knowledge, skills and behaviors, and attitudes.

The formative evaluation plan approach will be a quick-turnaround evaluation. The quick turnaround focuses on running the learning module for a short period, possibly a few weeks, and analyzing learner results.

The formative evaluation goal is to determine if the learner's perception of the value of privacy for digital identities through encryption has increased and if increased privacy awareness is additive to organizations with long-term technology, security, or business goals and objectives.

Audience

The intended audience of the evaluation results consists of decision-makers, including subject matter experts, business managers, course developers, and training leaders willing to assert whether the inclusion of the "Encryption of Digital Identities" learning module in learning and training roadmaps is beneficial.

Subject matter experts will validate the content and objectives, whereas the managers and educational leaders can evaluate if the course is standalone or coupled with existing methods and training. Course developers can assess the module's learning mechanisms and determine the effectiveness and diversity of objective and constructed-response and performance tests and evaluations built into the course.

Issues

Questions to be answered by the subject matter experts, business managers, course developers, and training leaders include the following:

- Was the module content well organized?
- What suggestions for you have for improving the module?
- Were there any incorrect assumptions made about learners' knowledge, skill, or attitude toward privacy, digital identities, etc.?
- Was the module engaging?
- Did the module trigger interest in privacy or cryptography?
- Do learners feel that the definitions, explanations, and examples provided were effective?
- Is the amount of time provided for each section adequate for most learning styles?
- What, if any, extraneous content can be removed from the module?
- Are each of the module learning objectives effectively addressed?

Resources

The table below lists the evaluation instruments and equipment required to for each evaluation phase:

- SME Notes Form
- PowerPoint enabled computer
- Observation Notes Form
- Interview Protocol
- Attitude Survey
- PowerPoint enabled computers (speakers and input peripherals such as a keyboard and trackpad/mouse)
- Learner Performance

Evaluation Schedule

Evaluation Stage	Purpose	Revision phase of Instructional Unit	Participants	Evaluation Instruments & Equipment
SME Review	Content validation, technical quality control	Prototype	2-3 cryptography or digital identity subject matter experts	• SME Notes Form • PowerPoint enabled computer
One-to-one evaluations	Learner impression	Prototype Revision 1	Individual Learner (possible 2-3 repetitions)	• Observation Notes Form • Interview Protocol • Attitude Survey • PowerPoint enabled computer
Small-group	SWOT Analysis	Prototype Revision 2	Learner groups of 3-5 individuals 1-2 Course Developer	• Observation Notes Form • Interview Protocol • Attitude Survey • PowerPoint enabled computer
Field Test	Delivery and implementation	Revised Instructional Unit	Larger learner group (3-5) individuals	• Attitude Survey • Learner Performance • PowerPoint enabled computer

Evidence

The evidence from the formative evaluation will collected during each evaluation phase will capture the following:

- Qualitative data
 - Classifying learners (e.g., technical, non-technical)
 - Sample size
 - 2-3 SMEs with cryptographic, digital privacy, and digital identity expertise
 - 8-13 learners of technical and non-technical backgrounds randomly chosen for one-to-one, small-group, and field test evaluation
 - Information objectivity
 - Course content referenced from peer reviewed materials
- Quantitative data

- Evaluation instruments will consist of SME Notes, Attitude surveys, interview protocol, and learner performance tests to quantify engagement, effectiveness, and effort required to meet or exceed the learning module objectives

Data-Gathering

SME Notes Form – Subject matter experts provide feedback after evaluating the course for content validity and reliability

Observation Notes Form – Course developers will provide visual and non-visual feedback after observing learners within the small-group, if needed, the field test and notate any actionable changes to improve the module.

Interview Protocol – Randomly selected learners (3 to 5) from the small-group and field test phases will complete evaluations to capture direct feedback on the module's impression and any suggestions that may improve the quality of the module.

Attitude Survey - Randomly selected learners (3 to 5) from the small-group and field test phases will complete evaluations to identify agreement, disagreement, or indifference to the unit's novelty, perceived inherent value, recommendation rating, clarity, rigor, and notable suggestions.

Learner Performance – All learners will complete the posttest during the small-group and field test to assess learning through objective tests, specifically multiple choice, true/false questions)

Analysis

Revisions made to the learning module are based on qualitative and quantitative analysis following the data-gathering process. An instructional designer revises the learning module after baselining results for each evaluation phase to determine if learning performance improvements or regressions occurred following revision.

Among quantitative analysis procedures, posttest scores, learner performance distributions, and measures of central tendency (mean, median, mode) are summarized. For qualitative analysis, comments and free form input are filtered to learner and observer sentiment and satisfaction.

Reporting

An analytical report is provided to all stakeholders of the learning module including but not limited to training leadership, instructional designers, and any other interested parties. The goal of the reporting is to help with the decision-making process of including, excluding, or supplementing the learning module to conform to new or existing learning goals.

In addition to the analytical report, access to the raw source data will be provided for data transparency.

Appendix

Subject Matter Expert Evaluation Notes Form

Thank you for participating in the formative evaluation plan for the “Encryption of Digital Identities”. Your feedback helps refine the content and design of the module. Please be as descriptive (and honest) as possible. Thank you again for your feedback and support!

Objective 1: Understand a “digital identity” and user’s expectation of security and privacy
How well did the learning module address the objective above?
☐ Excellent ☐ Well ☐ Average ☐ Not well ☐ Poor
What can be added / removed to meet the objective above?

Objective 2: Understand encryption’s role of providing privacy, authentication, integrity, nonrepudiation to digital identities
How well did the learning module address the objective above?
☐ Excellent ☐ Well ☐ Average ☐ Not well ☐ Poor
What can be added / removed to meet the objective above?

Objective 3: Understand the process of encryption and decryption
How well did the learning module address the objective above?
☐ Excellent ☐ Well ☐ Average ☐ Not well ☐ Poor
What can be added / removed to meet the objective above?

Objective 4: Distinguish between asymmetrical, symmetrical, and hash encryption

How well did the learning module address the objective above?

☐ Excellent ☐ Well ☐ Average ☐ Not well ☐ Poor

What can be added / removed to meet the objective above?

Observation Notes Form

Test group ☐ One-to-One

☐ Small-Group

☐ Field Test

Date _____

Start time _____

End time _____

[illegible]

Learner Interview Protocol

Thank you for taking the time to participate in this learning module and for participating in the interview evaluation. Your responses will be used to help improve the content, design, and layout of this unit. All responses will be kept confidential.

1. What were your first impressions of this learning module?

2. What was your initial reaction to the visual design of the learning module? What would you change?

3. What was your reaction to the format and layout of the learning module? What would you change?

4. Were the steps, practice, and feedback clear? If not, what was not clear about them?

5. Which, if any of the specific tasks were confusing or too difficult? Can you describe the task and the context?

6. What did you think of the images and screen shots? Did they support your understanding of the steps?

7. In what ways did the instructional unit (not) meet your expectations for learning the material?

Attitude Survey

Thank you for taking the time to participate in this learning module and for participating in the attitude survey. Your responses will be used to help improve the content, design, and layout of this unit. All responses will be kept confidential.

Question	Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree
The overall quality of the module is positive	☐	☐	☐	☐	☐
The content was engaging and interesting	☐	☐	☐	☐	☐
The content provides sufficient time to complete the review items	☐	☐	☐	☐	☐
I would recommend this module to others	☐	☐	☐	☐	☐
The media in the module enhanced the material presented	☐	☐	☐	☐	☐
This learning module sufficiently met my expectations for understanding digital identities, cryptography, and privacy	☐	☐	☐	☐	☐

Learner Performance / Posttest

Google form may be found here - <https://forms.gle/G5DVxyShLLqTBhMZ9>

Preview of the form is shown below

Encryption of Digital Identities Posttest

 John.I.mcswain@gmail.com (not shared) [Switch account](#) 

*** Required**

Select your group *

Choose ▼

 This is a required question

Private information that should not be made public should use a form of encryption. * 0 points

☐ True

☐ False

Hash functions can be reversed to reveal the original information or data * 0 points

☐ True

☐ False

What is ciphertext? * 0 points

☐ Incomprehensible and undecipherable data

☐ Original, readable state of information or data

Symmetrical encryption uses public and private key pairs * 0 points

☐ True

☐ False

Encryption is the process of converting ciphertext into plaintext. * 0 points

☐ True

☐ False

The strength of encryption is not affected by the length or complexity of the secret key. * 0 points

- ☐ True
- ☐ False

Integrity verification determines if a file or data has been changed or modified. * 0 points

- ☐ True
- ☐ False

Nonrepudiation is the existence of unforgeable evidence information was sent or received * 0 points

- ☐ True
- ☐ False

Authorization is the use of traits or characteristics (factors) to verify a digital identity * 0 points

- ☐ True
- ☐ False

Digital identities are collections of data about a subject that represent attributes, preferences, or traits. * 0 points

- ☐ True
- ☐ False

Submit

Clear form

Needs Assessment

Google Survey Link - <https://forms.gle/cvf9qGyL5DMVhHE56>

Preview of the form is shown below

Software App User Survey

This form collects anonymous information regarding users of software application and their understanding of encryption

 john.i.mcswain@gmail.com (not shared) [Switch account](#)  Draft restored

* Required

What is your current age range? *

- ☐ Under 12 years old
- ☐ 12-17 years old
- ☐ 18-24 years old
- ☒ 25-34 years old
- ☐ 35-44 years old
- ☐ 45-54 years old
- ☐ 55-64 years old
- ☐ 65-74 years old
- ☐ 75 years or older

What is the highest degree or level of school you have completed? If currently enrolled, highest degree received. *

- ☐ Nursery school to 8th grade
- ☐ Some high school, no diploma
- ☐ High school graduate, diploma or the equivalent (for example: GED)
- ☐ Some college credit, no degree
- ☐ Trade/technical/vocational training
- ☐ Associate degree
- ☐ Bachelor's degree
- ☐ Master's degree
- ☐ Professional degree
- ☐ Doctorate degree

Are you currently...? *

- ☐ Employed
- ☐ Self-employed
- ☐ Out of work and looking for work
- ☐ Out of work but not currently looking for work
- ☐ A homemaker
- ☐ A student
- ☐ Military
- ☐ Retired
- ☐ Volunteer

Do you use any of the web or desktop software applications listed below? *
Please select all that apply.

- ☐ Slack
- ☐ Zoom
- ☐ Grammarly
- ☐ Dropbox
- ☐ iCloud
- ☐ Microsoft OneDrive
- ☐ LinkedIn
- ☐ Facebook
- ☐ Google Chrome
- ☐ Mozilla Firefox
- ☐ Microsoft Edge
- ☐ Trello
- ☐ Gmail
- ☐ Microsoft Outlook
- ☐ DocuSign
- ☐ Other: _____

What is the purpose of using the software apps you selected above? *

- ☐ Business/Professional
- ☐ Personal
- ☐ Both
- ☐ Other: _____

How confident do you feel about your understanding of how web or desktop software applications work? *

- | | | | | | | |
|----------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|---------------|
| | 1 | 2 | 3 | 4 | 5 | |
| Very confident | ☐ | ☐ | ☐ | ☐ | ☐ | Not confident |

Do you consider digital privacy when choosing or using a software application? *

- ☐ Yes
- ☐ No
- ☐ Maybe
- ☐ I don't know

Have you ever been notified of a security incident/breach with any web or desktop applications you use? *

- ☐ Yes
- ☐ No
- ☐ Maybe
- ☐ I don't know

Do you know if any software applications you selected use encryption for data in use, in motion, or at rest? *

- ☐ Yes
- ☐ No
- ☐ Maybe
- ☐ I don't know

How confident do you feel about your understanding of the importance of encryption? *

	1	2	3	4	5	
Very confident	☐	☐	☐	☐	☐	Not confident

How confident do you feel about your understanding of how encryption works? *

	1	2	3	4	5	
Very confident	☐	☐	☐	☐	☐	Not confident

Do you take any extra steps to protect your digital identity? Please select all that apply. *

- ☐ VPN
- ☐ Password management (1Password, LastPass, etc.)
- ☐ Multi-factor Authentication (SMS/MMS, authentication apps, etc.)
- ☐ Monitor online activity (transactions, devices signed-in, etc.)
- ☐ Other: _____

Submit

Clear form

References

DevGlan. (n.d.). Online Tool for AES Encryption and Decryption. Devglan.
<https://www.devglan.com/online-tools/aes-encryption-decryption>

Jean-Philippe, A. (2018). *Serious Cryptography*. No Starch Press.

Morrison, G. R., Ross, S. J., Morrison, J. R., & Kalman, H. K. (2019). *Designing Effective Instruction* (8th ed.). Wiley.

Tadayoshi Kohno, Niels Ferguson, & Schneier, B. (2010). *Cryptography Engineering : Design Principles and Practical Applications*. Wiley Pub., Inc.

What is asymmetric cryptography? | NordVPN. (2020, February 3). Nordvpn.com.
<https://nordvpn.com/blog/what-is-asymmetric-cryptography/>

Windley, P. J. (2005). *Digital identity* (1st ed). O'Reilly.